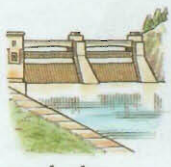
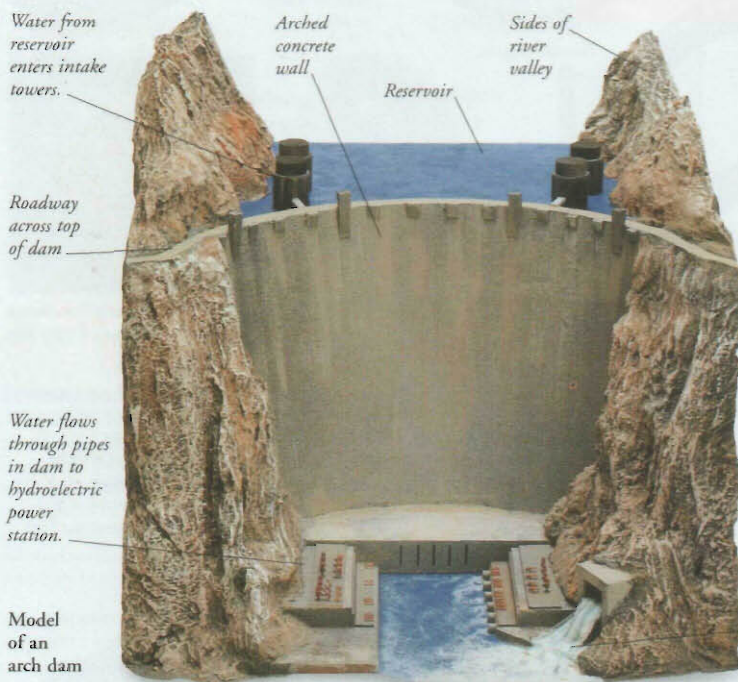


DAMS



IN MANY AREAS of the world, people rely on dams for their water and electricity supplies. A dam is a barrier that holds back water. The dam itself and the surrounding hills form a bowl in which water collects to form an artificial lake called a reservoir. Most dams are built across a river valley to catch the river's flow, but some dams create reservoirs into which water is pumped for storage. How strong a dam needs to be depends on the depth of the water in the reservoir. Some dams are enormous: the Grand Coulee Dam in the USA weighs nearly 10 million tonnes.



Environmental effects

A river dam and the reservoir it forms can harm the environment. Huge areas of countryside are drowned by the reservoir, and the dam disrupts the river's natural flow, affecting wildlife and irrigation downstream. A dam also prevents fish from moving freely up and down the river.



Tidal barrage

A barrage is a dam across a river estuary that generates hydroelectric power. The dam holds back the tide as it ebbs and flows. The water is forced through pipes inside the barrage, where it drives electricity generators.

La Rance barrage, France

Weir

A weir is a low river dam that controls the flow of water by creating a stretch of deeper water upstream. Deep water makes the river navigable for boats.



Weir in Middlesex, England

Types of dam

There are three main types of dam: arch dams, gravity dams, and buttress dams. The type of dam that engineers decide to build depends on the geography of the location. Factors affecting the decision include the width and depth of the river valley and the type of rock around the site.



Arch dam

An arch dam is built across the entrance to a narrow valley, so that the height of the dam is greater than its width. The dam's curved shape holds back water because it transfers the push of the water to the rock of the valley sides.



Buttress dam

A buttress dam is a huge concrete wall that leans into a reservoir of water. The wall is made up of concrete slabs that are supported on the downstream side of the dam by concrete projections known as buttresses.



Gravity dam

A gravity dam is a huge embankment of earth or rock. Leakage is prevented by a waterproof clay core or a concrete skin on the upstream side of the dam. The dam's immense weight prevents the water from pushing it over.

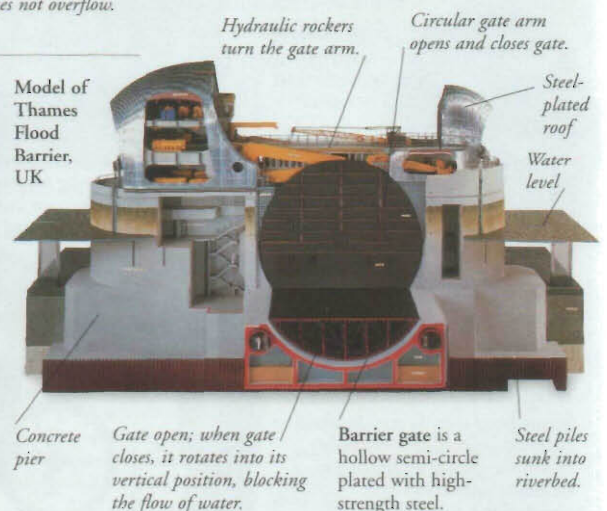
Anatomy of a dam

This model shows an arch dam that creates a reservoir for supplying water and electricity to nearby towns and cities. The dam is made of thin concrete strengthened by thousands of steel bars. Water flowing through pipes in the dam drives electricity generators in the hydroelectric power station at the foot of the dam.

Spillway lets excess water flow into river, so that dam does not overflow.

Flood control

On large rivers, dams help prevent flooding by holding back surges of flood water and releasing them downstream slowly. A flood barrier is a movable dam built across a tidal river. The barrier has gates that are usually open to allow the river to flow freely, but which can be closed when dangerously high tides threaten to surge upstream.



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